BRIEF 03 · DIGITAL SOLUTIONS & AI

# Predict mini-grid demand before building

**Focus region** West and East Africa | **Data** Public + partner

## MINI-CASE STUDY

*Amina is a mini-grid developer. Her last site was oversized, and the tariff needed to repay it priced out half the village. The site before that was undersized and browned out every evening. Her lenders want a demand forecast, but her only inputs are old census counts and short surveys where households guess at what they will use.*

Persona is illustrative, drawn from common conditions in the focus region.

## WHO IS IMPACTED AND WHY IT MATTERS

Villages waiting for power are hurt most: poor estimates lead to unaffordable tariffs, brownouts, or projects that never get financed. The World Bank estimates that reaching universal access by 2030 would take about 217,000 mini-grids serving 490 million people, at a cost of $127 billion [1]. Developers use demand estimates both to design systems and to secure financing, so forecasting errors ripple through the whole pipeline [1].

## EVIDENCE AT A GLANCE

| FINDING | WHAT IT MEANS FOR TEAMS | REF. |
|---|---|---|
| Most surveyed mini-grids had demand of 5–35 kWh per customer per month, a sevenfold difference in revenue expectations. | A forecast off by that range can make or break a project. | 1 |
| Developers use demand estimates to design mini-grids and to secure financing. | The model's output must be credible to lenders, not just engineers. | 1 |
| A household-only mini-grid runs at about a 22% load factor; adding daytime business use to reach 40% cut the cost of power by 25%. | Predicting productive use is as important as predicting household demand. | 2 |
| World Bank analysis found more than 177,000 settlements of 100–500 people and about 96,000 of 500–2,500 people suited to mini-grids. | There is a large, mappable universe of candidate sites. | 3 |
| Users in these areas spend about $5–$20 per month on candles, kerosene, batteries, and diesel. | Current energy spending is a usable signal of willingness to pay. | 3 |
| At current pace, only about 44,800 new mini-grids serving 80 million people would be built by 2030. | Faster, cheaper site assessment is part of closing the gap. | 4 |

## ROOT CAUSES

**Frame this as a forecasting and trust problem, not just a modeling exercise. Developers and lenders need estimates they can defend, including how demand will grow after connection.**

1. **Little data exists before connection.** Unelectrified villages have no meter history to learn from.
2. **Surveys mislead.** Households struggle to predict use of electricity they have never had.

3. **Operators do not share data.** Metered consumption sits in private company systems, so each developer starts from scratch.
4. **Demand varies widely and changes after connection.** Demand per customer ranges roughly sevenfold across sites [1], and grows as households buy appliances.
5. **Productive use is hard to predict.** Business loads drive load factor and cost of power [2], but depend on local markets and finance.
6. **Lenders discount uncertain forecasts.** Because forecasts underpin financing [1], uncertainty raises the cost of capital or shrinks systems.

## WHAT HAS BEEN TRIED

- Geospatial electrification planning using settlement, grid, and population layers (GRID3, GridFinder, WorldPop) [3]
- Portfolio-scale mini-grid programs and standardized designs promoted by the World Bank [4]
- Appliance financing and productive-use programs that raise load factor after connection [2]
- Remaining gap: open, validated models that predict demand growth at a specific site with a credible range

## CONSTRAINTS

- Must work where ground data is thin
- Show a range with confidence, not a single number
- Explain which inputs drive the estimate so a developer can trust it
- Run on a laptop or free cloud tier

## HOW EACH DISCIPLINE CONTRIBUTES

| DISCIPLINE | OWNS | QUESTIONS TO ANSWER | TOOLS & DATA |
|---|---|---|---|
| **Data science & AI** | Demand model, uncertainty ranges | Which features predict demand? How wide is the range? | Satellite data, building footprints, metered data |
| **Geospatial analysis** | Site features from satellite and maps | How do you count businesses from above? | Open Buildings, VIIRS, GridFinder, OpenStreetMap |
| **Power systems engineering** | Load profile and system size | What peak and daily profile follow from the estimate? | HOMER, NREL REopt |
| **Business & finance** | Tariff, revenue, bankability | How wrong can the forecast be before the project fails? | Mini-grid cost benchmarks, tariff data |
| **Community & design** | Local economy and appliance uptake | What will people buy first after connection? | Interviews, household surveys |

## DATA PACK

| DATASET | SOURCE | USE | ACCESS |
|---|---|---|---|
| **Building footprints** | Google Open Buildings; OpenStreetMap | Count and size structures | Public |
| **Nighttime lights** | NOAA VIIRS | Signal of existing activity | Public |
| **Grid lines** | GridFinder | Distance to grid | Public |
| **Settlements and services** | Energy Access Explorer | Schools, clinics, markets | Public |
| **Household energy surveys** | World Bank Multi-Tier Framework surveys | Link household traits to use | Public |
| **Metered consumption from live sites** | Mini-grid operator | Train and test the model | **Partner needed** |
| **Mini-grid cost and performance benchmarks** | ESMAP; AMDA benchmarking reports | Link demand to project finances | Public |
| **Electrification planning datasets** | Global Electrification Platform (World Bank / ESMAP) | Least-cost planning context | Public |

## DELIVERABLES

- Model or app that returns a demand estimate with a range for any location
- Demand growth curve for the first years after connection
- Validation against held-out metered sites
- Plain-language explanation of the main drivers a lender can review
- Proposal for how operators could share meter data safely

## SUCCESS METRICS

| | |
|---|---|
| **Impact · 25%** | Error versus actual metered demand on held-out sites |
| **Innovation · 25%** | New data sources or modeling methods that predict demand growth, not just day-one demand |
| **Sustainability · 25%** | Model maintainable with open data and periodic retraining; reduces oversized systems and wasted capital |
| **Feasibility · 15%** | Runs on public inputs alone, validated with partner data |
| **Scalability · 10%** | Works in more than one country with minimal retraining |

## STARTER QUESTIONS

1. Why do surveys over- or under-predict demand?
2. How does demand grow in the first two years after connection?
3. Which signals best predict businesses, not just households?

4. How wrong can an estimate be before it breaks the project's finances?
5. What would convince an operator to share meter data?

## REFERENCES

[1] ESMAP / World Bank (2022). Mini Grids for Half a Billion People: Market Outlook and Handbook for Decision Makers. https://www.esmap.org/sites/default/files/esmap-files/Mini%20Grids%20for%20Half%20a%20Billion%20People%20-%20OKR%20Download%20Version_compressed_MKA.pdf

[2] ESMAP / World Bank. Mini Grids for Half a Billion People (full report, load factor and LCOE analysis). https://sdgs.un.org/sites/default/files/2021-05/Mini%20Grids%20For%20Half%20A%20Billion%20People%20-%20Market%20Outlook%20And%20Handbook%20For%20Decision%20Makers.pdf

[3] Digital Energy summary of ESMAP analysis: Mini-grids for half a billion people. https://digital-energy.eu/en/studies/esmap-mini-grids-half-milion-people

[4] REGlobal. Mini Grids for Half a Billion People: World Bank Handbook. https://reglobal.org/mini-grids-for-half-a-billion-people-world-bank-handbook/

*Sources accessed September 2026. Figures from older studies are cited as historical context.*

**EVERY TEAM SUBMITS**

A working prototype or model, a 3-minute pitch, and a one-page impact estimate using the brief's success metrics.

**JUDGED ON**

Impact (25%), innovation (25%), sustainability (25%), technical feasibility (15%), and scalability (10%).